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Module	**Data Visualisation**

Customer Behaviour Analysis Report

Introduction

The company offers six shipping methods, but management does not know which of these options deliver the greatest business value. Although each method differs in speed, convenience, and operational cost, the organisation lacks insight into how customers behave when selecting shipping types. This analysis examines customer count, average spend, and estimated revenue across all shipping methods to identify high-volume and high-value patterns. The goal is to determine which two shipping methods the company should prioritise for investment in the next quarter.

Findings

The visualisations reveal clear behavioural differences across shipping types. Express and Standard Shipping attract the highest customer volumes, with approximately 907 and 937 customers respectively. These two methods also generate the highest average spend (USD273 and USD260), making them strong high-volume, high-value options. Their estimated revenues are significantly higher than all other methods.

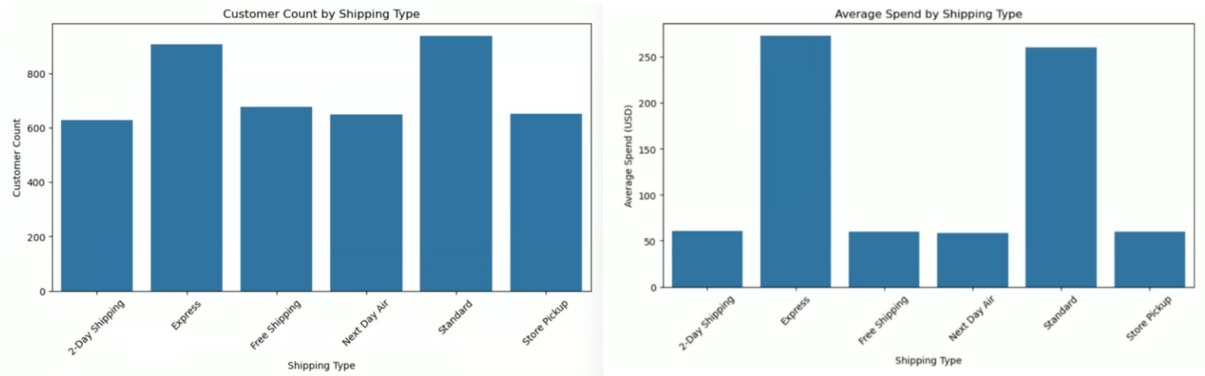

Fig.1. Bar charts for Customer Count and Average Spend provided simple comparisons across categories.

In contrast, Free Shipping, Store Pickup, 2-Day Shipping, and Next Day Air attract similar customer volumes (around 627–650 each) but are associated with much lower average spend, typically between USD 58 and USD 61. These methods represent high-volume but low-value behaviour, suggesting they appeal to more price-sensitive customers.

Fig. 2. A grouped Bar chart showing the Customer Count vs Average Spend by Shipping Type

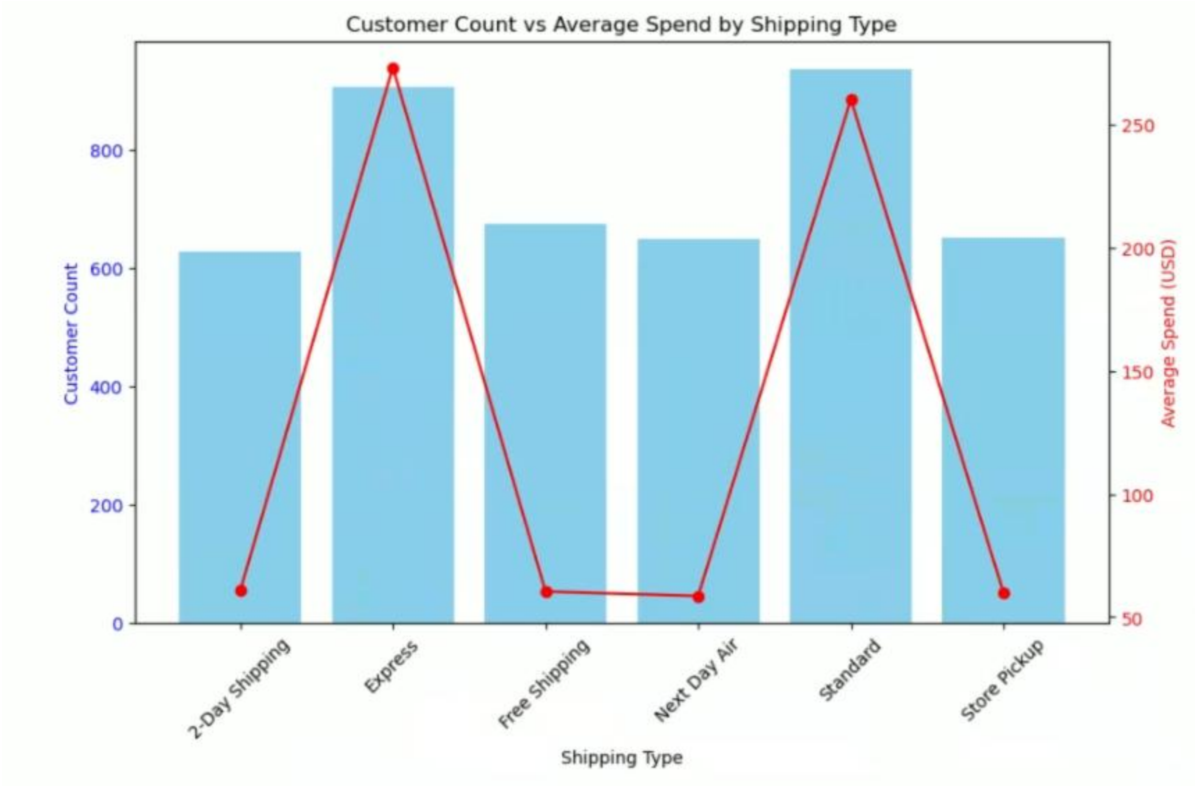

Fig. 3: A dual-axis chart for Customer Count vs Average Spend by Shipping Type

The dual-axis chart shows that faster delivery does not consistently lead to higher spend. While Express customers spend considerably more, other fast options such as 2-Day Shipping and Next Day Air show some of the lowest average spend values. This indicates that the relationship between delivery speed and spending is not straightforward and should be interpreted cautiously.

Evaluation of Visual Choices

Four visualisations were used to explore the data:

1. Bar charts for Customer Count and Average Spend (Fig. 1) provided simple comparisons across categories.
2. A dual-axis chart (Fig. 3) was chosen to compare volume and value simultaneously. This chart was particularly effective because the two metrics operate on very different scales, and separate y-axes allowed both patterns to be visible.
3. A grouped bar chart (Fig. 2) was included for completeness, but its shared y-axis compressed the spend values, making differences harder to interpret.
4. A revenue bar chart (Fig. 4) clearly highlighted which shipping methods generated the highest total value.

Overall, the dual-axis chart offered the strongest insight because it directly addressed the business question.

Recommendation

Based on customer behaviour and revenue performance, the company should prioritise Express Shipping and Standard Shipping. These two methods attract the largest customer bases and generate the highest average spend and estimated revenue. Investing in these options is likely to deliver the strongest business impact in the next quarter.

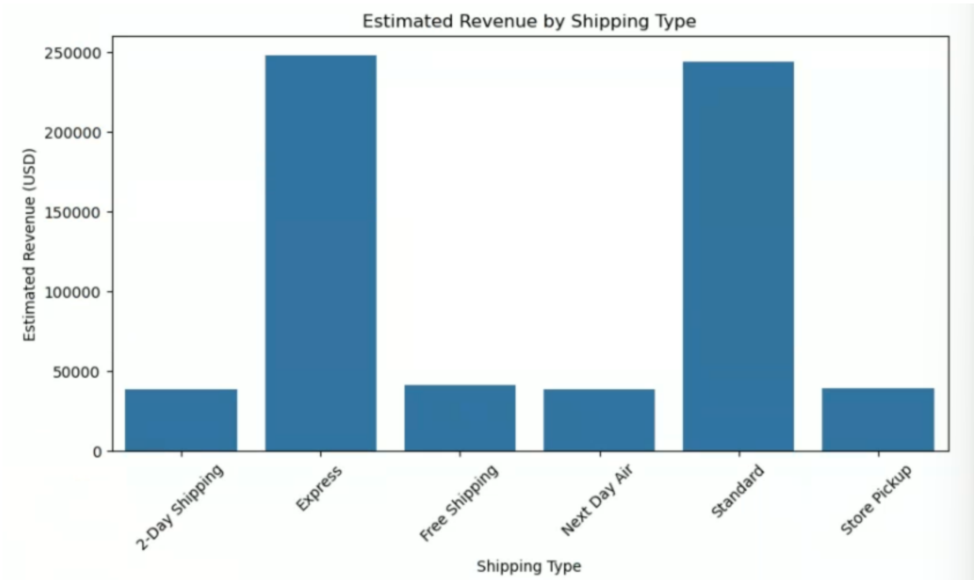

Fig. 4. Estimated Revenue by Shipping Type

Finally, additional data such as shipping cost, profit per shipping type, customer return rates would significantly improve the analysis. These variables would help explain whether those choices are profitable or operationally efficient.